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Urbanization restructures small mammal communities without reducing species richness

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Abstract

Urban wildlife persistence is often interpreted through island biogeography theory (IBT), in which habitat fragments function as islands within an inhospitable matrix. In cities, however, the matrix is heterogeneous and may strongly influence dispersal and habitat conditions. Small mammals may be especially sensitive to urbanization because many species avoid impervious surfaces and experience multiple urban environmental stressors. We surveyed small mammal communities at 23 sites along an urban–rural gradient northwest of Atlanta, Georgia, USA, and quantified spatial, landscape, and socioeconomic variables to evaluate alternative drivers of community structure. We captured 1,123 individuals representing 13 species in 14,720 trap-nights. Species richness did not differ among rural, suburban, and urban sites, but population densities of White-Footed Mice (*Peromyscus leucopus*) and Eastern Chipmunks (*Tamias striatus*) were higher in suburban and urban sites. Multivariate analyses revealed clear shifts in community composition along the gradient despite stable richness. Beta diversity partitioning showed that rural–nonrural differences were driven primarily by species turnover, whereas suburban and urban sites differed mainly through nestedness. Variables associated with IBT, including fragment size and isolation, did not explain variation in species richness or most population metrics. Instead, several responses were linked to landscape context: *P. leucopus* and generalist rodents increased with developed land cover, whereas *T. striatus* occurrence increased with impervious surface and human population density. Socioeconomic variables had limited explanatory power. Together, these results indicate that urbanization restructures small mammal communities through species turnover and demographic responses of generalist taxa, with landscape context and matrix composition outweighing classic island-biogeography processes.

Introduction

Urbanization has increased dramatically over the past century and is expected to continue expanding for several decades (United Nations 2024). As cities grow, natural landscapes are converted to human-dominated land uses, resulting in habitat loss, fragmentation, and altered environmental conditions (Johnson and Karels 2015; Simkin et al. 2022). These changes affect not only human communities but also the wildlife capable of persisting within cities (McKinney 2008).

Small mammals—terrestrial rodents and eulipotyphlans with body mass <2 kg—are common inhabitants of urban landscapes, yet their responses to urbanization remain incompletely understood, with studies reporting inconsistent patterns with respect to species richness, community composition, and the mechanisms driving responses (Lokatis and Jeschke 2022). These species are of particular interest because of their ecological roles and their frequent interactions with humans, including disease transmission, vehicle collisions, and use of anthropogenic food sources (McCleery et al. 2021; Rimbach et al. 2023). Many small mammals show limited willingness to cross impervious surfaces (McGregor et al. 2008; Brehme et al. 2013), making it possible to delineate habitat fragments and examine how landscape structure influences their distribution. In addition, well-characterized life histories allow small mammal assemblages to be evaluated both taxonomically (e.g., abundance or species richness) and functionally through ecological strategies or trait-based approaches (Cadotte et al. 2011).

Urbanization can alter community composition through processes such as biotic homogenization, in which similar environmental conditions across cities favor disturbance-tolerant generalist species while reducing the prevalence of habitat specialists (McKinney 2006). In some systems, these changes involve replacement of native species by synanthropic rodents such as the House Mouse (*Mus musculus*) and Brown Rat (*Rattus norvegicus*), whereas in others urbanization primarily promotes native generalist species such as White-Footed Mouse (*Peromyscus leucopus*) capable of exploiting anthropogenic environments (Cavia et al. 2009, Rimbach et al. 2025).

One framework commonly used to interpret wildlife responses to habitat fragmentation is island biogeography theory (IBT; MacArthur and Wilson 1967). IBT predicts that species richness should increase with patch area and decrease with isolation from source populations. Urban habitat fragments may resemble ecological "islands," with impervious surfaces and built structures functioning as the surrounding matrix (Johnson and Karels 2015; Olejniczak et al. 2018). Several studies have reported positive relationships between fragment area and small mammal richness in urban landscapes (e.g., Pardini et al. 2005; Johnson and Karels 2015), although evidence for the role of isolation is less consistent.

In addition to fragment size and isolation, other characteristics of habitat remnants may influence small mammal persistence. Fragment geometry can affect edge exposure and resource availability within patches, producing species-specific responses to habitat boundaries (Pardini et al. 2005). Similarly, fragment age and disturbance history may influence ecological succession and extinction probabilities within remnants (Bolger et al. 2000). These factors may therefore affect community composition independently of classic island biogeography predictions.

Landscape context surrounding habitat fragments may also shape small mammal communities. The composition and permeability of the surrounding matrix can influence movement, immigration, and persistence of species within habitat patches (Brehme et al. 2013; Fletcher et al. 2024). In addition, socioeconomic factors may indirectly affect wildlife communities because neighborhood wealth and population density can influence land-use policies, vegetation management, and habitat structure (Pickett and Pearl 2001). The "luxury effect," in which wealthier neighborhoods support greater biodiversity, has been documented for several taxa including bats (Li and Wilkins 2014), inconsistently in medium and large mammals (Magle et al. 2021), and biota generally (Leong et al. 2018), although its relevance to small mammals remain poorly understood.

Environmental conditions within habitat fragments may further influence small mammal assemblages. Habitat structure, vegetation cover, and resource availability are known to affect the abundance and persistence of many small mammal species (Green and Wilkins 2014). Such local habitat characteristics may interact with broader landscape processes to shape community composition in urbanizing regions.

Questions and hypotheses

To evaluate how urbanization shapes small mammal communities, we examined assemblages across rural, suburban, and urban landscapes in the metropolitan region of Atlanta, Georgia, USA—a heavily forested North American city with tree canopy cover exceeding 50% of municipal land area (Merry et al. 2014). We tested four hypotheses concerning how urbanization may influence small mammal communities. These hypotheses are not necessarily mutually exclusive, and multiple processes may operate simultaneously in shaping small mammal communities along urbanization gradients.

Hypothesis 1 was that urbanization alters small mammal community structure. We tested the specific predictions that species richness, community composition, and population densities would differ among rural, suburban, and urban landscapes. Hypothesis 2 was that small mammal responses to urbanization are driven by island biogeography–like processes, because if habitat remnants function as ecological islands within an inhospitable urban matrix, fragment size and isolation should influence immigration and extinction dynamics. Our prediction was that species richness and population densities would vary with patch characteristics such as fragment area, isolation, age, and perimeter attributes.

Hypothesis 3 was that landscape context drives community responses independent of patch geometry, because composition and permeability of the surrounding matrix may influence movement and persistence of species within habitat remnants. We predicted that variation in community composition and species abundances would be associated with surrounding land cover and other landscape characteristics rather than fragment size or isolation alone. Finally, hypothesis 4 was that local socioeconomic context influences small mammal communities because human population density and household income can affect land-use practices. We predicted that small mammal diversity and abundance would vary with local socioeconomic conditions.

Methods

We surveyed small mammal communities at 23 sites along an urban–rural gradient extending northwest from Atlanta, Georgia, USA (Casement 2024). Spatial, environmental, and socioeconomic variables were measured within and surrounding each site and used to test the study hypotheses using ordination and regression-based analyses.

Study area

We sampled 23 sites along an urban–rural gradient extending roughly along U.S. Interstate 75 from Fulton County to Bartow County, Georgia, USA (Fig. 1). Sites were classified as urban (n = 7), suburban (n = 8), or rural/natural (n = 8) based on satellite imagery and county property tax records. The regional climate is humid subtropical, with mean summer daily high temperature of 26.8 °C and mean annual precipitation of 126 cm. Most sites occurred within the

Piedmont Level III ecoregion, with several near the Blue Ridge or Ridge and Valley boundaries. All sites were dominated by successional pine or mixed pine–hardwood forest, the primary native habitat of the region (U.S. Environmental Protection Agency 2013).

Small mammal trapping

Small mammals were captured using Sherman live traps (model LFATDG, 9 × 9 × 23 cm; H.B. Sherman Traps, Tallahassee, Florida, USA) baited with a mixture of crimped oats and commercial bird seed. Traps were arranged in transects of 20 stations spaced 10 m apart with two traps per station (Green et al. 2024; McCleery et al. 2021). Sites were sampled for four consecutive nights between May and August 2023, yielding 640 trap-nights per site. Sites were treated as the sampling unit for all analyses and contained four transects (most sites) or five transects (Kennesaw State University Field Station).

Captured animals were weighed, measured, identified to species, assessed for sex and reproductive condition, marked with ear tags (model 1005-1, National Band and Tag Company, Newport, Kentucky, USA), and released at the point of capture. Traps containing animals were replaced with clean traps, and all traps were sanitized with a 10% bleach solution between sites (Yunger and Randa 1999). All procedures followed the American Society of Mammalogists guidelines for the use of wild mammals in research (Sikes et al. 2016) and were approved by the Kennesaw State University Institutional Animal Care and Use Committee and performed under a Georgia Department of Natural Resources Scientific Collecting Permit issued to N. S. Green.

Spatial and environmental data collection

Analyses focused on landscape- and site-level predictors rather than within-site microhabitat variables. Sites were selected to represent similar habitat types across the urban–rural gradient, minimizing systematic differences in local habitat structure. Microhabitat measurements were collected at the transect scale and were therefore not aggregated to the site level.

Geospatial predictor variables were calculated using geographic information systems (GIS). Site boundaries were digitized in ArcGIS Pro 3.1.0 (Esri, Redlands, California, USA). Site area was calculated directly from these polygons. Fragment isolation was defined as the minimum distance from a site to the nearest area of natural land cover. Fragment age was estimated as the most recent year each site was connected to surrounding habitat based on historical aerial photographs (imagery source: https://dlg.usg.edu/collection/gyca_gaphind, date accessed 29 April 2024). Fragment shape complexity S was quantified as $S = (0.25P)/\sqrt{A}$, where P is the site perimeter and A is the area, given that P and $\sqrt{A}$ have the same units. Elevation (m) was calculated as the mean elevation within each site using the National Elevation Dataset (US Geological Survey 2019). Perimeter imperviousness was measured as the percentage of each site boundary bordered by impervious surfaces.

Landscape variables were calculated within 100-m buffers surrounding each site. This buffer size was chosen based on movement distances ($\approx 25 \pm 53$ m day^{-1}) and home-range sizes of *P. leucopus*, the most frequently captured species, and to avoid overlap among sites. Human population density was obtained from the WorldPop project using unconstrained UN-adjusted population counts for 2020 at approximately 1-ha resolution (www.worldpop.org, date accessed 21 April 2025). Land cover and impervious surface variables were derived from the National Land Cover Dataset (NLCD) for 2020 (https://www.mrlc.gov, accessed 21 April 2025). Socioeconomic variables were obtained from the U.S. Census Bureau American Community Survey (ACS) for 2021 (https://www.census.gov/programs-surveys/acs/, accessed 21 April 2025). For each site buffer we calculated mean values of human population density, impervious surface cover, tree canopy cover, and other land cover categories (grass, wetlands, cropland, and water). Median household income was calculated as the area-weighted mean of census polygons intersecting each buffer. Full geospatial processing details are provided in Casement (2024).

Population density and biodiversity metrics

Analyses included all rodents and eulipotyphlans captured during sampling, but excluded Virginia Opossums (*Didelphis virginiana*). Because Southern Short-Tailed Shrews (*Blarina carolinensis*) could not be individually marked, we assumed that nine of the ten captures represented unique individuals based on capture dates and locations (N.S. Green, personal observation). Small mammal population densities were calculated as the minimum number known alive (MNKA) divided by the species-specific area sampled by all transects at each site (Green and Wilkins 2014). We used MNKA because recapture frequencies were insufficient to support more complex capture-recapture estimators. Sampling area for each transect was estimated using species-specific movement distances, measured as mean maximum distance moved (MMDM; Hammond and Anthony 2006). We estimated MMDM values from our capture-recapture data for *P. leucopus*, Cotton Mouse (*Peromyscus gossypinus*), Hispid Cotton Rat (*Sigmodon hispidus*), and Eastern Chipmunk (*Tamias striatus*). For other species, movement distances were obtained from published studies. Sampling areas were calculated from transect length and species-specific movement distances following Green and Wilkins (2014). Additional details and species-specific movement values are provided in Casement (2024). Biodiversity was measured as species richness because numerical dominance by *P. leucopus* limited the interpretability of diversity indices.

Ecological strategy groups

We identified ecological strategy groups (ESGs) of functionally similar species using life-history traits of small mammals potentially occurring at our sites, as a secondary way of summarizing functional similarities between species. Species were grouped using *k*-means clustering (Legendre and Legendre 2012; de Bello et al. 2021) based on traits obtained from the Phylacine database (Faurby et al. 2018). These traits included variables related to body size, diet, habitat association, and pace of life (e.g., lifespan and reproductive frequency). Clustering was conducted for all small mammal species known from Georgia rather than only captured species so that the resulting groups reflected the broader regional fauna. Six ESGs were identified; we focused on ESG2 (habitat specialist muroids) and ESG6 (habitat generalist muroids) because they represented contrasting ecological strategies. ESG6 included primarily *P. leucopus* with contributions from *M. musculus* and *P. gossypinus*, whereas ESG2 included *S. hispidus*, Pine Vole (*Microtus pinetorum*), Eastern Harvest Mouse (*Reithrodontomys humulis*), and Golden Mouse (*Ochrotomys nuttalli*). ESG densities were calculated similarly to species densities using the combined MNKA and abundance-weighted mean of MMDM of species within each group. Full details of the clustering procedure are provided in Casement (2024).

Statistical analyses

Community composition (Hypothesis 1).

We first tested for spatial autocorrelation in response variables using Mantel tests and detected none ($P > 0.05$). Sites were therefore treated as statistically independent. Community composition was visualized using nonmetric multidimensional scaling (NMDS) based on Bray-Curtis dissimilarities. Differences in community composition among rural, suburban, and urban sites were tested using permutational multivariate analysis of variance (PERMANOVA).

Homogeneity of multivariate dispersion among groups was evaluated using the *betadisper* procedure to ensure PERMANOVA results were not driven by differences in dispersion. Environmental vectors were fitted to NMDS ordinations using the *envfit* function in package *vegan* (Oksanen et al. 2025) to explore correlations between ordination structure and spatial variables. Only significant vectors ($P < 0.05$) were displayed, and results were used for interpretation rather than hypothesis testing. To further characterize community differences, we partitioned presence–absence beta diversity (β_{SOR}) into turnover (β_{SIM}) and nestedness (β_{SNE}) components (Baselga 2012). Kruskal–Wallis tests were used to evaluate differences in species richness and selected population densities among urbanization categories, followed by Dunn's tests for post-hoc comparisons when appropriate.

Landscape variables (Hypotheses 2, 3, 4).

To test effects of landscape processes, generalized linear models (GLMs) were used to examine relationships between univariate response variables and patch or landscape attributes. Species richness was modeled using Poisson GLMs with a log link. Population densities of *P. leucopus* and ESG6 were modeled using linear regression. Presence of *S. hispidus*, *T. striatus*, and ESG2 were analyzed using logistic regression. Significance of linear models was assessed using ANOVA and significance of Poisson and binomial GLMs using likelihood ratio tests.

To test IBT-like processes (Hypothesis 2), we modeled univariate responses using patch attributes including area, shape complexity, age, isolation distance, and perimeter imperviousness. For hypothesis 3, landscape predictors included mean tree canopy cover, impervious cover, forest cover, developed cover, and open land cover within 100-m buffers around each site. For hypothesis 4, socioeconomic predictors included human population density, median household income, and the percentage of households below the poverty line within the

100-m buffers. Each predictor was tested individually to avoid multicollinearity and because our objective was to evaluate *a priori* hypotheses with a limited sample size.

All analyses were performed using R version 4.5.1 (R Core Team 2025). We used package *vegan* version 2.6-10 for all ordinations and multivariate analyses (Oksanen et al. 2025); package *dunn.test* version 1.3.6 for Dunn's tests (Dinno 2024); and package *betapart* version 1.6.1 for beta diversity analysis (Baselga et al. 2025). All geospatial calculations were performed in R using packages *terra* version 1.7-65 (Hijmans 2023) and *sp* version 1.5-0 (Bivand et al., 2013; Pebesma et al., 2005).

Results

Capture summary

We captured 1,123 small mammals representing 13 species in 14,720 trap-nights across 23 sites (7.6 captures per 100 trap-nights). *Peromyscus leucopus* dominated the captures, accounting for 852 individuals (76%) at 20 sites. Other common species included *S. hispidus* (114 captures at 7 sites) and *T. striatus* (48 captures at 10 sites). Two non-native species were detected: *R. norvegicus* (13 captures at 4 sites) and *M. musculus* (14 captures at 1 site). Additional species captured in small numbers were *P. gossypinus* (24), *M. pinetorum* (22), *R. humulis* (15), *B. carolinensis* (11), *S. carolinensis* (3), *G. volans* (3), *O. nuttalli* (2), and *D. virginiana* (2).

Hypothesis 1: Effects of urbanization on small mammal communities

Univariate analyses

Across all species, neither species richness (KW χ^2_2 = 0.9384, P = 0.6255), ESG richness (KW χ^2_2 = 2.6150, P = 0.2705), *S. hispidus* population density, (KW χ^2_2 = 4.8463, P = 0.0886), generalist muroids (ESG6; KW χ^2_2 = 4.0041, P = 0.1351), nor specialist muroids (ESG2; KW χ^2_2 = 0.3201, P = 0.8521) were significantly different between site types (urban, suburban, or rural).

Population density differed between site types for *P. leucopus* (KW $\chi^2 = 6.2204$, $P = 0.0446$) and *T. striatus* (KW $\chi^2 = 8.3565$, $P = 0.0153$). Dunn's post hoc tests showed that for *P. leucopus*, density was smaller in rural sites than suburban sites (Dunn's $Z = -2.3614$, $P = 0.0091$) or urban sites (Dunn's $Z = -1.8434$, $P = 0.0326$), and for *T. striatus* that density was smaller in rural sites than in suburban (Dunn's $Z = -2.7270$, $P = 0.0032$) or urban (Dunn's $Z = -2.1571$, $P = 0.0155$) sites (**Error! Reference source not found.** 2).

Multivariate analyses

NMDS ordination of species densities produced a two-dimensional solution (stress = 0.142; Fig. 3). *Sigmodon hispidus, R. humulis*, and *P. gossypinus* were associated with rural sites, while *S. carolinensis*, *G. volans*, *R. norvegicus*, and *T. striatus* were associated with suburban and urban sites. *Peromyscus leucopus* was not clearly associated with any site type. Urban and suburban sites were associated with greater elevation, site age, and shape complexity, while rural sites were surrounded by more open land cover (i.e., land cover that was neither forest nor developed).

Homogeneity of multivariate dispersion did not differ among site types (betadisper permutation test $F_{2,20} = 2.025$, $P = 0.158$), supporting interpretation of PERMANOVA results as differences in group centroids. Urbanization status explained 17.8% of variation in species densities (PERMANOVA $F_{2,20} = 2.1634$, $P = 0.024$). Post-hoc tests showed that this difference was driven by a difference between rural and suburban sites (PERMANOVA $F_{1,14} = 2.878$, $r^2 = 0.170$, $P = 0.021$) and between rural and nonrural sites (PERMANOVA $F_{1,21} = 3.190$, $r^2 = 0.132$, $P = 0.011$; $P > 0.05$ in all other comparisons).

Beta diversity partitioning

Total presence-absence β diversity (β_{SOR}) was similar across contrasts, but the relative contributions of turnover (β_{SIM}) and nestedness (β_{SNE}) differed among site types (Table 1). Partitioning of presence–absence β diversity indicated that differences between rural and nonrural sites were driven primarily by species turnover, whereas suburban and urban sites showed little evidence of species replacement and differed mainly through nestedness (Table 1). Species turnover between suburban and urban sites was typically zero, while contrasts involving rural sites often reflected replacement of 1 to 3 species rather than simple species losses in nonrural sites compared to rural sites.

Hypothesis 2: Island-like processes on small mammals

No IBT-like process had a significant effect on small mammal species richness. Population densities of *P. leucopus* ($F_{1,21}$ = 10.17, P = 0.004, r^2 = 0.294) and ESG6 ($F_{1,21}$ = 8.78, P = 0.007, r^2 = 0.261) were negatively affected by %-impervious perimeter, which also negatively affected *T. striatus* detection probability (likelihood ratio test: χ^2_1 = 8.085, P = 0.004; McFadden pseudo-r^2 = 0.257; AUC = 0.792; Fig. 4a). Detection probability of *S. hispidus* was negatively affected by fragment age (likelihood ratio test: χ^2_1 = 12.206, P < 0.001; McFadden pseudo-r^2 = 0.411; AUC = 0.892; Fig. 4d). Site area, shape complexity and site isolation were not associated with any response variable. For hypotheses 2, 3, and 4, full model parameter summaries and test statistics are presented in Supplementary Data SD1.

Hypotheses 3 and 4: Local landscape and socioeconomic factors

Landscape-level spatial factors

Population density of *P. leucopus* ($F_{1,21}$ = 10.26, P = 0.004, r^2 = 0.328) and ESG6 ($F_{1,21}$ = 7.557, P = 0.012, r^2 = 0.265), and occurrence of *T. striatus* (likelihood ratio test: χ^2_1 = 10.295, P =

0.001; McFadden pseudo-r^2 = 0.327; AUC = 0.838), increased with increasing developed cover within 100 m of the site (Fig. 4b). *Peromyscus leucopus* population density declined both with increasing forest cover ($F_{1,21}$ = 4.412, P = 0.048, r^2 = 0.174) and with increasing non-forest, non-developed ("open") land cover ($F_{1,21}$ = 4.404, P = 0.048, r^2 = 0.173; Fig. 4c;), whereas ESG6 abundance declined with forest cover ($F_{1,21}$ = 4.876, P = 0.038, r^2 = 0.150). *Tamias striatus* occurrence increased with increasing local impervious surface (likelihood ratio test: χ^2_1 = 7.932, P = 0.005; McFadden pseudo-r^2 = 0.251; AUC = 0.788).

Socioeconomic factors

Occurrence of *T. striatus* increased with local human population density (likelihood ratio test: χ^2_1 = 6.438, P = 0.011; McFadden pseudo-r^2 = 0.204; AUC = 0.877); no other small mammal endpoint was significantly predicted by human population density, and none by local median household income or poverty rate.

Discussion

Urbanization altered small mammal communities in our study system primarily through changes in species composition and species-specific demographic responses rather than through large declines in species richness. To interpret these patterns, we evaluated four hypotheses regarding how urbanization might influence small mammal communities and discuss our results below in the context of each hypothesis.

Hypothesis 1: Small mammal community structure is affected by urbanization

We found modest evidence that urbanization influenced small mammal community structure. Species richness did not differ among rural, suburban, and urban sites, but community composition shifted and population densities of *P. leucopus* and *T, striatus* increased in suburban and urban landscapes. Several muroid habitat specialists (*R. humulis*, *S. hispidus*, and *M.*

pinetorum) were strongly associated with rural sites and were largely absent from suburban and urban sites. In contrast, the generalist *P. leucopus* occurred across all sites and reached higher densities in developed landscapes. Although the ordination showed separation among site types, variation in *P. leucopus* abundance did not closely follow the primary urban–rural gradient, suggesting that factors other than urbanization alone may influence its distribution.

Many studies of urban gradients report declining richness and increasing dominance by generalist or synanthropic species toward urban cores (Cavia et al. 2009; Łopucki and Kitowski 2017; Magura et al. 2021; Rimbach et al. 2025). However, our results highlight a more nuanced pattern in which community composition shifts while total richness remains relatively stable. Similar patterns have been reported elsewhere, where urbanization restructures assemblages without necessarily reducing species counts. For example, Ofori et al. (2018) and Ofori et al. (2025) documented rural–urban differences in community composition despite relatively stable richness, and Rimbach et al. (2025) found that urban landscapes maintained comparable richness while selectively favoring generalist taxa.

The absence of a strong richness decline in our study contrasts with studies that report substantial species loss along urban gradients (e.g., Łopucki and Kitowski 2017). Two factors likely explain this difference. First, many studies reporting steep declines include heavily built environments where only a few commensal species persist. Our sampling instead focused on habitat fragments embedded within urban landscapes, which likely constrained the species pool to taxa capable of using these greenspaces. Previous work has shown that within-patch habitat quality can partially offset surrounding urbanization, allowing remnant habitats to support relatively diverse small mammal assemblages (Garden et al. 2007). Second, several Nearctic rodents, particularly *P. leucopus*, are highly adaptable habitat generalists capable of persisting across developed landscapes (Munshi-South and Karchenko 2010). Systems dominated by broadly tolerant native species may therefore exhibit compositional reshuffling rather than strong declines in richness.

Patterns of beta diversity in our study further support this interpretation. Comparisons involving rural sites showed greater species replacement, whereas suburban and urban sites shared largely similar species pools. This suggests that suburban and urban landscapes retain a subset of disturbance-tolerant species while rural sites maintain additional habitat specialists. These patterns are consistent with processes often associated with urban biotic homogenization, in which urbanization filters communities toward a smaller set of disturbance-tolerant taxa (Garba et al. 2014).

Interestingly, we did not observe strong replacement of native species by non-native *Mus* and *Rattus* species, which has been reported in some cities (Cavia et al. 2009; Ofori et al. 2025). Instead, community change appeared to involve modest shifts toward more generalist native species. This pattern aligns with broader syntheses suggesting that urbanization often promotes disturbance-tolerant generalists rather than simply replacing native faunas with invasive taxa (Magura et al. 2021; Wang et al. 2025). Persistence in urban habitat patches may therefore depend on opportunistic life-history traits that allow species to exploit novel environments associated with suburban and urban development.

Hypothesis 2: Small mammal responses affected by island-like processes

Classic island biogeography theory (IBT) predicts that species richness should increase with patch size and decrease with isolation. However, many urban studies report weak or inconsistent support for these predictions. For example, Ekernas and Mertes (2006) found no relationship between patch size and diversity, while Johnson and Karels (2015) showed that fragment area influenced richness primarily through its effects on habitat heterogeneity rather than through area alone. Instead, habitat quality and matrix permeability often appear to be stronger drivers of small mammal responses (Garden et al. 2007; Cassel et al. 2020).

Our findings are consistent with this broader literature. Neither fragment size nor isolation distance affected any small mammal endpoint in our study. These results suggest that classical IBT may provide limited predictive power in urban landscapes. Unlike oceanic islands, urban matrices are heterogeneous and not uniformly hostile. Residential yards, riparian corridors, and other semi-natural features can facilitate movement among habitat fragments even where impervious surfaces and roads pose barriers.

Several studies have also shown that within-patch vegetation structure and microhabitat conditions frequently explain more variation in small mammal communities than patch geometry alone (Pardini et al. 2005; Garden et al. 2007). Our results support this interpretation, as responses were more closely associated with matrix characteristics such as impervious surface than with fragment area or isolation.

Although classical island biogeography processes were weak in our system, dispersal limitation imposed by the urban matrix may still influence community structure. Roads and other paved surfaces can impede movement for many small mammals, including *Peromyscus* and *Tamias* (Brehme et al. 2013; McGregor et al. 2008). Previous studies have documented reduced dispersal and gene flow for *P. leucopus* in urban landscapes, suggesting that even mobile species may experience functional isolation (Munshi-South and Karchenko 2010; Morton 2025). Rather than fragment size or distance alone, the permeability of the urban matrix may therefore play a larger role in determining connectivity among habitat patches.

In the absence of strong support for classical IBT, some aspects of neutral biogeography theory may provide a complementary framework for interpreting small mammal responses to urbanization. Neutral theory emphasizes the roles of dispersal limitation and demographic stochasticity in shaping community structure (Hubbell 2001). Although empirical tests generally indicate that neutral processes alone cannot fully explain community assembly, they may contribute to patterns observed in fragmented or dispersal-limited systems. Urban landscapes may amplify these dynamics by increasing stochastic extinctions and reducing connectivity among habitat patches. Although we did not explicitly test predictions of neutral theory, the weak effects of fragment size and isolation observed here are broadly consistent with a system in which dispersal limitation within an urban matrix interacts with species-specific habitat associations to shape community structure.

Hypotheses 3 and 4: Landscape context and socioeconomic factors

Landscape context.

Both island biogeography and neutral frameworks assume that dispersal among sites is mediated by the surrounding landscape matrix. Consistent with this expectation, developed land cover within 100 m of sites was the most consistent predictor of small mammal responses in our study. Developed cover was positively associated with *P. leucopus*, *T. striatus*, and the generalist muroid assemblage (ESG6), whereas both forested and open land cover were negatively associated with *P. leucopus*. These results suggest that certain generalist species may benefit from heterogeneously developed landscapes, particularly where development creates edge habitat and potential anthropogenic food subsidies.

Although the generalist muroid group (ESG6) was numerically dominated by *P. leucopus*, the broader assemblage also declined with increasing forest cover, suggesting that generalist species in this system may preferentially exploit edge-rich or developed habitats rather than extensive continuous forest (Cassel et al. 2020). Numerous studies have documented the success of *Peromyscus* species in urban environments, where increased edge habitat and anthropogenic resources may partially compensate for the loss of continuous forest (Ekernas and Mertes 2006; Munshi-South and Kharchenko 2010).

Our findings also complement those of Larson and Sander (2025), who examined small mammal responses to urbanization in prairie-dominated landscapes. Similar to our results, they reported positive associations between anthropogenic land cover and small mammal diversity and abundance but also observed species-specific responses to imperviousness. Differences between their findings and ours likely reflect contrasts in regional habitat context. Their study area was dominated by prairie ecosystems, whereas our system is primarily forested. In prairie-dominated landscapes, increased tree cover and development may increase habitat heterogeneity, whereas in forest-dominated landscapes development may create structurally heterogenous edge environments that benefit generalist species relative to open habitats. These comparisons highlight the importance of interpreting urbanization effects within the context of regional habitat structure.

Environmental conditions within sites may also play an important role. Reviews of mammal responses to landscape change suggest that local habitat characteristics often predict species richness more strongly than broad landscape variables (Garden et al. 2007; Cassel et al. 2020). Urban tree canopy cover in Atlanta exceeds 50% of land area in some estimates (Merry et al. 2014). Extensive urban forest cover may therefore reduce ecological contrasts between suburban and urban environments and contribute to broadly similar mammal assemblages across the urban matrix.

Socioeconomic factors.

We hypothesized that small mammal population densities would increase with human population density because anthropogenic food sources such as garbage could subsidize populations of species able to exploit them. However, only *P. leucopus* and *T. striatus* showed positive associations with human population density. These species may benefit from anthropogenic resources or behavioral flexibility that allows them to persist in developed environments.

In contrast, the typically human-associated rodents *R. norvegicus* and *M. musculus* showed no relationship with urbanization metrics, likely because of their low capture rates. These species were detected only in suburban and urban sites, suggesting that they are present in the urban landscape but may be poorly sampled using Sherman traps.

Socioeconomic variables often serve as proxies for broader urban environmental conditions, including impervious surface, building density, and waste availability. However, these variables had limited explanatory power for small mammal communities in our study. We also found little evidence for the "luxury effect," in which biodiversity increases in wealthier neighborhoods. Although this pattern has been documented for plants and some animal groups, evidence for mammals remains mixed (Magle et al. 2021).

Some studies have documented "poverty effects" in which inadequate sanitation and waste management promote rodent infestations (Himsworth et al. 2013; Zeppelini et al. 2021). Because captures of *R. norvegicus* and *M. musculus* were rare in our dataset, we were unable to test these mechanisms directly.

Conclusions

We expected small mammal diversity to be highest in rural landscapes, lowest in urban landscapes, and intermediate in suburban areas. Instead, many differences occurred primarily between rural and nonrural sites. Rural habitats supported several specialist species that were rare or absent in developed landscapes, whereas suburban and urban sites shared a similar set of disturbance-tolerant taxa.

However, developed landscapes did not uniformly reduce biodiversity. Some species and functional groups were more associated with suburban or urban environments, and population densities of several species were higher in these landscapes. Suburban environments may provide a mixture of habitat features that allow some species to exploit both natural and anthropogenic resources.

Our study had several limitations. The relatively small number of sites limited our ability to apply occupancy modeling, and we therefore relied on correlation analyses. Low capture rates also required greater trapping effort per site than in some comparable studies. In addition, several species known to occur at our sites, including squirrels and flying squirrels, are poorly sampled using Sherman traps.

Despite these limitations, our results highlight the importance of landscape context in shaping small mammal communities in urbanizing environments. Rather than following classical island biogeography predictions, community responses were driven primarily by habitat filtering and species-specific responses to developed land cover. Urbanization in this system appears to reorganize community composition through demographic responses of tolerant species rather than through uniform species loss.

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Supplementary Data

Supplementary data SD1: This document contains the tables of summary outputs of linear and generalized linear models of species richness, population density of *P. leucopus*, occurrence probability *S. hispidus*, occurrence probability of *T. striatus*, occurrence probability of muroid habitat specialists (ESG2), and population density of muroid habitat generalists (ESG6).

Data availability

Data and R scripts to support the final analysis are publicly available on GitHub (https://github.com/greenquanteco/Casement-et-al-spatial-ecology).

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Tables

Table 1. Summary of total beta diversity (β_{SOR}) calculated as Sørensen dissimilarity between sites partitioned into turnover (β_{SIM}) and nestedness (β_{SNE}) components. Values shown are median of site-site comparisons with interquartile ranges (IQR) in parentheses.

Site-type contrast	Total (β_{SOR})	Turnover (β_{SIM})	Nestedness (β_{SNE})
Rural vs. suburban	0.67 (0.50, 0.76)	0.50 (0.00, 0.50)	0.17 (0.00, 0.33)
Rural vs. urban	0.63 (0.42, 0.83)	0.50 (0.00, 0.81)	0.09 (0.00, 0.33)
Suburban vs. urban	0.50 (0.33, 0.60)	0.00 (0.00, 0.50)	0.21 (0.08, 0.43)
Rural vs. nonrural	0.67 (0.50, 0.78)	0.50 (0.00, 0.69)	0.10 (0.00, 0.33)

691 **Figures**

692 **Fig. 1**. Map of study sites along an urban-to-rural gradient in Bartow, Cobb, Fulton, and DeKalb
693 counties, Georgia. Shading shows human population density in 2020 (WorldPop 2020). Inset
694 shows state of Georgia (grey) within the continental United States of America. The study region
695 is indicated by the red square.

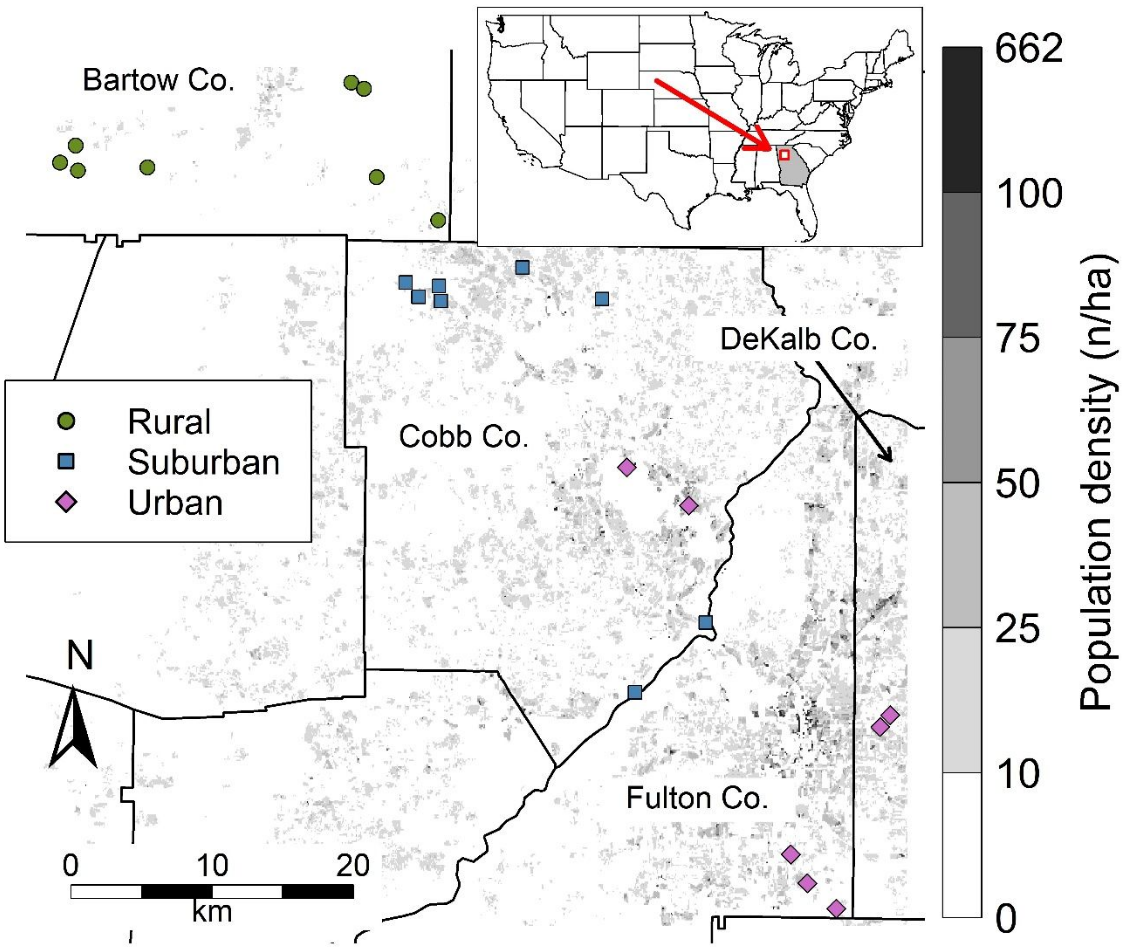

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Fig. 2. a. Small mammal species richness did not differ between rural, suburban, and urban sites. b. White-Footed Mouse (*Peromyscus leucopus*) population density was greater in suburban and urban sites than rural sites. c. Hispid Cotton Rat (*Sigmodon hispidus*) population density did not differ between site types. d. Eastern chipmunk (*Tamias striatus*) population density was greater in suburban and urban sites than rural sites

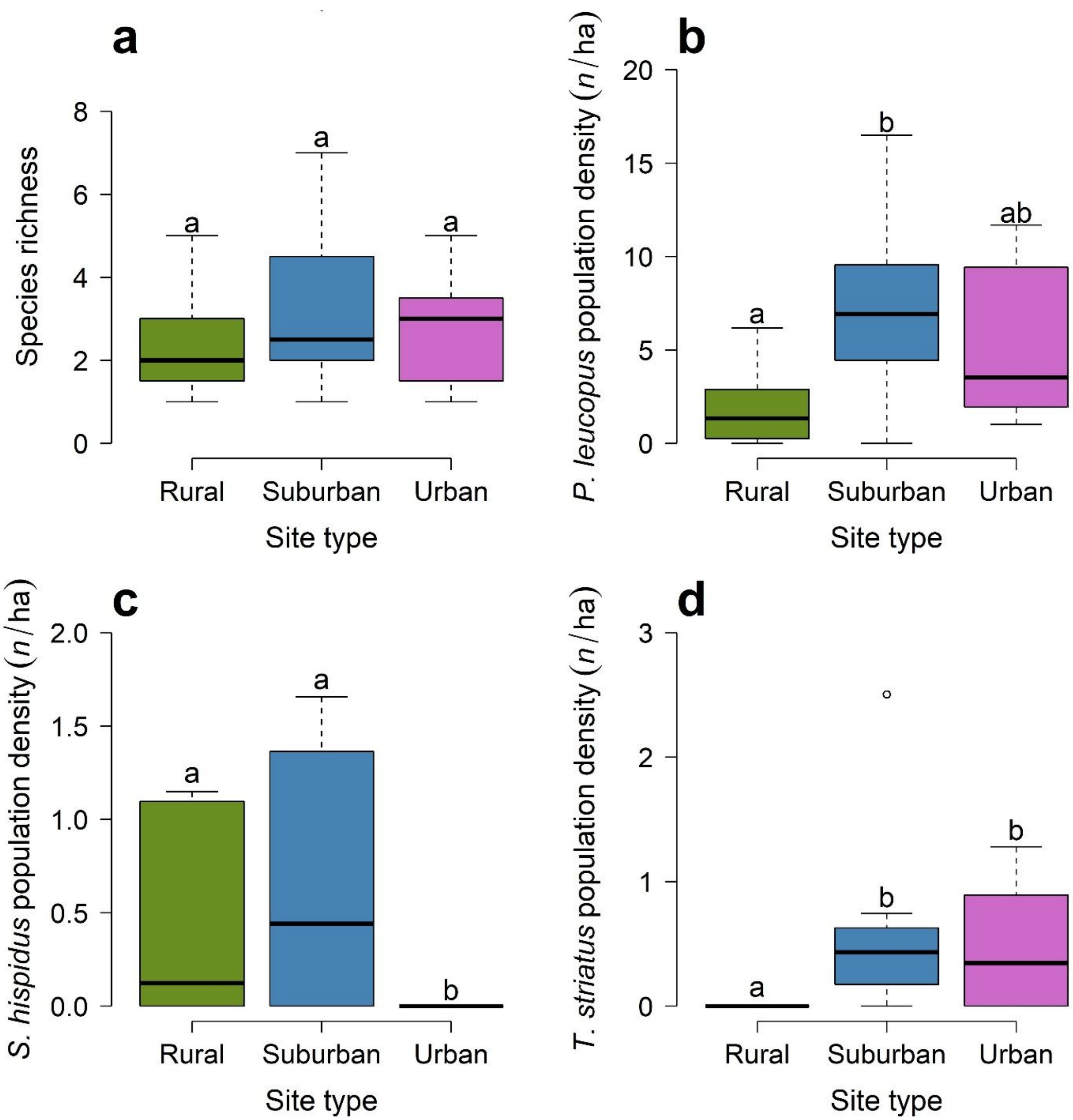


Fig. 3. Two-dimensional nonmetric multidimensional scaling (NMDS) solution showing community differences between rural, suburban, and urban sites (stress = 0.142). Points show sites; black vectors show species associations with the NMDS-space; blue vectors show environmental variables fitted to the NMDS-space. Permutational multivariate analysis of variance (PERMANOVA) indicated that rural sites and non-rural sites hosted different small mammal assemblages. Species abbreviations: Bc = *Blarina carolinensis*; Gv = *Glaucomys volans*; Mm = *Mus musculus*; Mp = *Microtus pinetorum*; On = *Ochrotomys nuttalli*; Pg = *Peromyscus gossypinus*; Pl = *P. leucopus*; Rh = *Reithrodontomys humilis*; Rn = *Rattus norvegicus*; Sc = *Sciurus carolinensis*; Sh = *Sigmodon hispidus*; Ts = *Tamias striatus*. Environmental variables are site age, mean elevation (“Elev.”), shape complexity (“Shape”), and amount of open land cover within 100 m of the site boundaries.

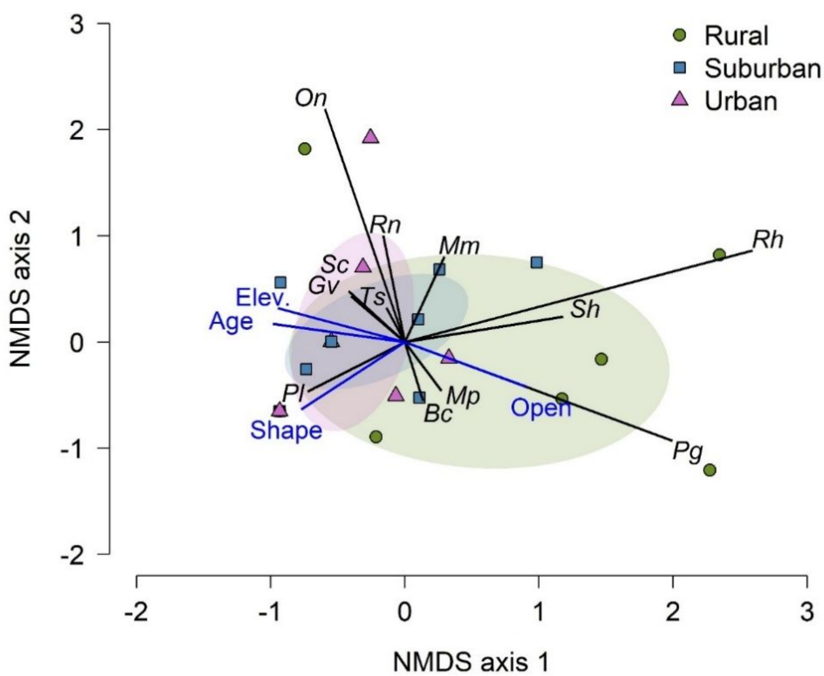
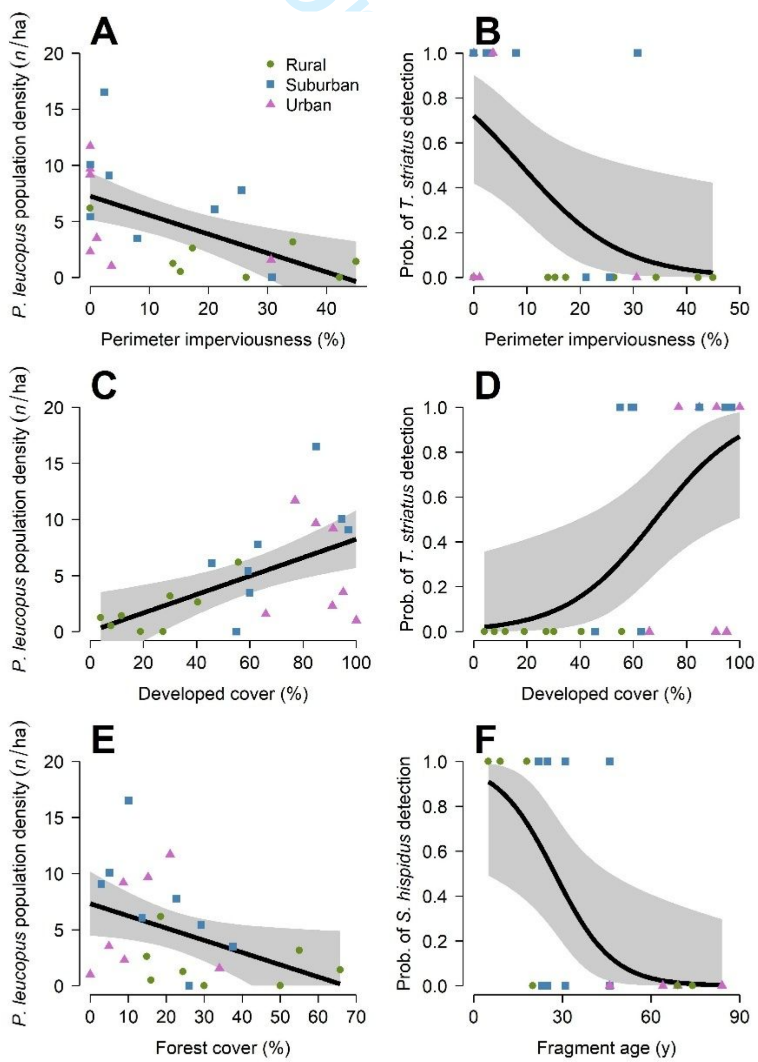


Fig. 4. Population density of White-Footed Mouse (*Peromyscus leucopus*) and occurrence probability of Eastern Chipmunk (*Tamias striatus*) decreased with increasing site perimeter imperviousness (A,B). Abundance of *P. leucopus* and occurrence of *T. striatus* increased with increasing developed cover within 100 m of the site (C,D). *Peromyscus leucopus* abundance decreased with increasing forest cover (E). Hispid Cotton Rat (*Sigmodon hispidus*) occurrence probability decreased with increasing fragment age (F). Lines and shaded areas represent model predictions and 95% confidence intervals (CI); predicted population densities are truncated at 0 for display purposes only.



Supplementary Data SD1. Model output summaries for models of univariate response variables

Supplement to manuscript entitled, "Urbanization restructures small mammal communities without reducing species richness" by B. Casement, L. Lopez, K. Nestor, and N.S. Green.

This document contains the summary outputs of generalized linear models and linear models of species richness (Table SD1.1), population density of white-footed mice (*Peromyscus leucopus*; Table SD1.2), probability of occurrence of hispid cotton rat (*Sigmodon hispidus*; Table SD1.3), probability of occurrence of eastern chipmunk (*Tamias striatus*; Table SD1.4), probability of occurrence of ecological strategy group 2 (muroid habitat specialists; ESG2; Table SD1.5), and population density of ESG6 (muroid habitat generalists; Table SD1.6). All models reported in a table have the same response variable, and each model has a single predictor. Refer to manuscript or Casement (2024) for a complete description of all variables.

Species richness

Table SD1.1. Outputs of Poisson generalized linear models (GLM) of small mammal species richness using 3 groups of predictor variables related to island biogeography theory (IBT-like) for hypothesis 2; local land cover for hypothesis 3; and socioeconomic factors for hypothesis 4. Each predictor was tested in isolation to avoid cross-correlation. Estimate ± SE is the point estimate and standard error of that estimate; *z* is the test statistic; and *P* is the *P*-value.

Group	Model	Term	Estimate ± SE	*z*	*P*
IBT-like	1	Intercept	1.194 ± 0.203	5.869	<0.001
		Site area	-0.005 ± 0.005	-1.006	0.314
	2	Intercept	1.689 ± 0.683	2.472	0.013
		Shape complexity	-0.522 ± 0.534	-0.979	0.328
	3	Intercept	1.158 ± 0.247	4.689	<0.001
		Site age	-0.003 ± 0.005	-0.618	0.536
	4	Intercept	0.970 ± 0.149	6.512	<0.001
		Site isolation	0.001 ± 0.001	0.698	0.485
	5	Intercept	1.149 ± 0.164	6.993	<0.001
		% Imp. Perim.	-0.010 ± 0.009	-1.090	0.276
Land cover	6	Intercept	0.953 ± 0.414	2.306	0.021
		Tree cover (mean)	0.002 ± 0.010	0.178	0.859
	7	Intercept	0.935 ± 0.227	4.116	<0.001
		Impervious cover	0.004 ± 0.009	0.477	0.633
	8	Intercept	1.169 ± 0.205	5.691	<0.001
		Forest cover (%)	-0.007 ± 0.008	-0.855	0.393

	9	Intercept	0.783 ± 0.292	2.681	0.007
		Developed cover (%)	0.004 ± 0.004	0.938	0.349
	10	Intercept	0.992 ± 0.142	6.981	<0.001
		Open cover (%)	0.004 ± 0.008	0.500	0.617
Socioeconomic	11	Intercept	0.916 ± 0.173	5.300	<0.001
		Population density	0.020 ± 0.021	0.958	0.338
	12	Intercept	0.799 ± 0.264	3.023	0.003
		Median household income	<0.001 ± <0.001	0.996	0.319
	13	Intercept	1.053 ± 0.179	5.866	<0.001
		Poverty rate	-0.003 ± 0.012	-0.224	0.823

White-footed mouse (*Peromyscus leucopus*) population density

Table SD1.2. Outputs of linear regression models of white footed mouse (*Peromyscus leucopus*) population density using 3 groups of predictor variables related to island biogeography theory (IBT-like) for hypothesis 2; local land cover for hypothesis 3; and socioeconomic factors for hypothesis 4. Each predictor was tested in isolation to avoid cross-correlation. Estimate ± SE is the point estimate and standard error of that estimate; *t* is the test statistic; and *P* is the *P*-value. Models with significant predictors are bolded for emphasis.

Group	Model	Term	Estimate ± SE	*t*	*P*
IBT-like	1	Intercept	4.166 ± 1.473	2.827	0.010
		Site area	0.019 ± 0.030	0.638	0.530
	2	Intercept	-3.828 ± 4.579	-0.836	0.413
		Shape complexity	6.760 ± 3.485	1.939	0.066
	3	Intercept	3.232 ± 1.889	1.711	0.102
		Site age	0.040 ± 0.040	1.008	0.325
	4	Intercept	4.887 ± 1.118	4.371	<0.001
		Site isolation	<0.001 ± 0.007	0.002	0.998
	5	**Intercept**	7.247 ± 1.077	6.730	<0.001
		% Imp. Perim.	-0.169 ± 0.053	-3.189	0.004
Land cover	6	Intercept	3.084 ± 3.123	0.988	0.335
		Tree cover (mean)	0.047 ± 0.077	0.606	0.551
	7	Intercept	2.995 ± 1.623	1.845	0.079
		Impervious cover	0.095 ± 0.067	1.409	0.173
	8	**Intercept**	**7.323 ± 1.447**	**5.059**	**<0.001**

		Forest cover (%)	**-0.109 ± 0.052**	**-2.100**	**0.048**
	9	**Intercept**	**0.027 ± 1.707**	**0.016**	**0.988**
		Developed cover (%)	**0.082 ± 0.026**	**3.203**	**0.004**
	10	**Intercept**	**5.811 ± 0.972**	**5.978**	**<0.001**
		Open cover (%)	**-0.123 ± 0.059**	**-2.099**	**0.048**
Socioeconomic	11	Intercept	3.555 ± 1.203	2.956	0.008
		Population density	0.269 ± 0.162	1.662	0.111
	12	Intercept	1.569 ± 1.919	0.818	0.423
		Median household income	<0.001 ± <0.001	1.946	0.065
	13	Intercept	4.464 ± 1.364	3.272	0.004
		Poverty rate	0.038 ± 0.087	0.432	0.670

Hispid cotton rat (*Sigmodon hispidus*) detection probability

Table SD1.3. Outputs of logistic regression models of hispid cotton rat (*Sigmodon hispidus*) occurrence using 3 groups of predictor variables related to island biogeography theory (IBT-like) for hypothesis 2; local land cover for hypothesis 3; and socioeconomic factors for hypothesis 4. Each predictor was tested in isolation to avoid cross-correlation. Estimate ± SE is the point estimate and standard error of that estimate; *z* is the test statistic; and *P* is the *P*-value. Models with significant predictors are bolded for emphasis.

Group	Model	Term	Estimate ± SE	*z*	*P*
IBT-like	1	Intercept	-0.906 ± 0.689	-1.316	0.188
		Site area	0.007 ± 0.014	0.531	0.596
	2	Intercept	1.061 ± 2.441	0.435	0.664
		Shape complexity	-1.325 ± 1.905	-0.696	0.487
	3	**Intercept**	**2.818 ± 1.389**	**2.029**	**0.042**
		Site age	**-0.103 ± 0.044**	**-2.327**	**0.020**
	4	Intercept	-0.868 ± 0.529	-1.642	0.101
		Site isolation	0.003 ± 0.003	0.863	0.388
	5	Intercept	-1.305 ± 0.679	-1.922	0.055
		% Imp. Perim.	0.045 ± 0.031	1.437	0.151
Land cover	6	Intercept	1.547 ± 1.633	0.948	0.343
		Tree cover (mean)	-0.059 ± 0.044	-1.339	0.181
	7	Intercept	0.117 ± 0.780	0.151	0.880
		Impervious cover	-0.040 ± 0.036	-1.104	0.270
	8	Intercept	-0.659 ± 0.732	-0.900	0.368

		Forest cover (%)	0.001 ± 0.026	0.052	0.959
	9	Intercept	0.359 ± 0.936	0.383	0.702
		Developed cover (%)	-0.017 ± 0.015	-1.162	0.245
	10	Intercept	-1.417 ± 0.589	-2.408	0.016
		Open cover (%)	0.129 ± 0.074	1.732	0.083
Socioeconomic	11	Intercept	-0.097 ± 0.586	-0.166	0.868
		Population density	-0.125 ± 0.104	-1.196	0.232
	12	Intercept	-1.169 ± 0.965	-1.211	0.226
		Median household income	<0.001 ± <0.001	0.636	0.525
	13	Intercept	0.352 ± 0.739	0.476	0.634
		Poverty rate	-0.108 ± 0.078	-1.397	0.163

Eastern chipmunk (*Tamias striatus*) detection probability

Table SD1.4. Outputs of logistic regression models of eastern chipmunk (*Tamias striatus*) occurrence using 3 groups of predictor variables related to island biogeography theory (IBT-like) for hypothesis 2; local land cover for hypothesis 3; and socioeconomic factors for hypothesis 4. Each predictor was tested in isolation to avoid cross-correlation. Estimate ± SE is the point estimate and standard error of that estimate; *z* is the test statistic; and *P* is the *P*-value. Models with significant predictors are bolded for emphasis.

Group	Model	Term	Estimate ± SE	*z*	*P*
IBT-like	1	Intercept	0.404 ± 0.772	0.524	0.601
		Site area	-0.019 ± 0.020	-0.968	0.333
	2	Intercept	-0.646 ± 2.191	-0.295	0.768
		Shape complexity	0.297 ± 1.665	0.179	0.858
	3	Intercept	-0.876 ± 0.885	-0.990	0.322
		Site age	0.015 ± 0.018	0.800	0.424
	4	Intercept	-0.694 ± 0.515	-1.348	0.178
		Site isolation	0.006 ± 0.004	1.392	0.164
	5	**Intercept**	**0.946 ± 0.650**	**1.455**	**0.146**
		% Imp. Perim.	**-0.106 ± 0.047**	**-2.244**	**0.025**
Land cover	6	Intercept	-0.577 ± 1.395	-0.414	0.679
		Tree cover (mean)	0.008 ± 0.034	0.237	0.812
	7	**Intercept**	**-2.040 ± 1.013**	**-2.014**	**0.044**
		Impervious cover	**0.086 ± 0.043**	**2.026**	**0.043**
	8	Intercept	0.857 ± 0.791	1.085	0.278

		Forest cover (%)	-0.054 ± 0.034	-1.578	0.115
	9	**Intercept**	**-4.083 ± 1.753**	**-2.329**	**0.020**
		Developed cover (%)	**0.060 ± 0.025**	**2.434**	**0.015**
	10	Intercept	0.607 ± 0.607	0.999	0.318
		Open cover (%)	-0.821 ± 0.792	-1.036	0.300
Socioeconomic	**11**	**Intercept**	**-1.358 ± 0.670**	**-2.028**	**0.043**
		Population density	**0.233 ± 0.114**	**2.046**	**0.041**
	12	Intercept	-2.457 ± 1.423	-1.726	0.084
		Median household income	<0.001 ± <0.001	1.561	0.119
	13	Intercept	-0.154 ± 0.606	-0.254	0.799
		Poverty rate	-0.010 ± 0.040	-0.247	0.805

Ecological strategy group 2 (habitat specialists) detection probability

Table SD1.5. Outputs of logistic regression models of muroid habitat specialist (ecological strategy group or ESG 2) occurrence using 3 groups of predictor variables related to island biogeography theory (IBT-like) for hypothesis 2; local land cover for hypothesis 3; and socioeconomic factors for hypothesis 4. Each predictor was tested in isolation to avoid cross-correlation. ESG2 included hispid cotton rat (*Sigmodon hispidus*), eastern harvest mouse (*Reithrodontomys fulvescens*), pine vole (*Microtus pinetorum*), and golden mouse (*Ochrotomys nuttalli*). Estimate ± SE is the point estimate and standard error of that estimate; *z* is the test statistic; and *p* is the p-value. Models with significant predictors are bolded for emphasis.

Group	Model	Term	Estimate ± SE	*z*	*P*
IBT-like	1	Intercept	1.236 ± 0.755	1.637	0.116
		Site area	0.005 ± 0.016	0.330	0.745
	2	Intercept	3.394 ± 2.492	1.362	0.188
		Shape complexity	-1.525 ± 1.897	-0.804	0.430
	3	Intercept	2.413 ± 0.952	2.533	0.019
		Site age	-0.024 ± 0.020	-1.189	0.248
	4	Intercept	1.572 ± 0.566	2.780	0.011
		Site isolation	-0.002 ± 0.004	-0.491	0.628
	5	Intercept	1.664 ± 0.663	2.510	0.020
		% Imp. Perim.	-0.017 ± 0.033	-0.521	0.608
Land cover	6	Intercept	0.554 ± 1.590	0.349	0.731
		Tree cover (mean)	0.023 ± 0.039	0.576	0.571
	7	Intercept	2.151 ± 0.843	2.552	0.019

		Impervious cover	-0.036 ± 0.035	-1.037	0.311
	8	Intercept	1.433 ± 0.810	1.769	0.091
		Forest cover (%)	>-0.001 ± 0.029	-0.009	0.993
	9	Intercept	2.117 ± 1.046	2.024	0.056
		Developed cover (%)	-0.012 ± 0.016	-0.742	0.466
	10	Intercept	1.068 ± 0.516	2.072	0.051
		Open cover (%)	0.048 ± 0.031	1.537	0.139
Socioeconomic	11	Intercept	1.451 ± 0.651	2.230	0.037
		Population density	-0.005 ± 0.087	-0.055	0.956
	12	Intercept	0.493 ± 1.035	0.476	0.639
		Median household income	<0.001 ± <0.001	1.015	0.322
	13	Intercept	1.519 ± 0.697	2.180	0.041
		Poverty rate	-0.008 ± 0.045	-0.183	0.857

Ecological strategy group 6 (habitat generalists) population density

Table SD1.6. Outputs of linear regression models of muroid habitat generalist (ecological strategy group or ESG 6) population density using 3 groups of predictor variables related to island biogeography theory (IBT-like) for hypothesis 2; local land cover for hypothesis 3; and socioeconomic factors for hypothesis 4. Each predictor was tested in isolation to avoid cross-correlation. ESG6 included white footed mouse (*Peromyscus leucopus*), cotton mouse (*P. gossypinus*), and house mouse (*Mus musculus*). Estimate ± SE is the point estimate and standard error of that estimate; *t* is the test statistic; and *p* is the p-value. Models with significant predictors are bolded for emphasis.

Group	Model	Term	Estimate ± SE	*t*	*P*
IBT-like	1	Intercept	4.597 ± 1.444	3.184	0.004
		Site area	0.019 ± 0.030	0.652	0.522
	2	Intercept	-2.212 ± 4.578	-0.483	0.634
		Shape complexity	5.841 ± 3.485	1.676	0.109
	3	Intercept	4.574 ± 1.887	2.424	0.024
		Site age	0.018 ± 0.040	0.454	0.654
	4	Intercept	5.293 ± 1.096	4.829	<0.001
		Site isolation	<0.001 ± 0.007	0.046	0.964
	5	**Intercept**	**7.519 ± 1.080**	**6.961**	**<0.001**
		% Imp. Perim.	**-0.158 ± 0.053**	**-2.964**	**0.007**
Land cover	6	Intercept	4.608 ± 3.084	1.494	0.150
		Tree cover (mean)	0.018 ± 0.076	0.242	0.811
	7	Intercept	3.762 ± 1.614	2.332	0.030

		Impervious cover	0.078 ± 0.067	1.167	0.256
	8	**Intercept**	**7.887 ± 1.396**	**5.652**	**<0.001**
		Forest cover (%)	**-0.115 ± 0.050**	**-2.297**	**0.032**
	9	**Intercept**	**1.040 ± 1.751**	**0.594**	**0.559**
		Developed cover (%)	**0.072 ± 0.026**	**2.749**	**0.012**
	10	Intercept	5.856 ± 1.016	5.764	<0.001
		Open cover (%)	-0.072 ± 0.061	-1.167	0.256
Socioeconomic	11	Intercept	4.191 ± 1.199	3.497	0.002
		Population density	0.228 ± 0.161	1.413	0.172
	12	Intercept	1.919 ± 1.865	1.029	0.315
		Mean household income	<0.001 ± <0.001	2.050	0.053
	13	Intercept	5.110 ± 1.342	3.808	0.001
		Poverty rate	0.019 ± 0.086	0.218	0.830